Student ID_____ Student Name_____

IS221: Web Applications Development

School of Information Technology, Engineering, Mathematics and Physics

Final Examination
Semester 1 2023

Face-To-Face and Online Mode

Duration of Exam: 3 hours + 10 minutes

Reading Time: 10 minutes

Writing Time: 3 hours

INSTRUCTIONS

1. This examination has two sections:
   a. Section A: Coding Questions (40 marks)
   b. Section B: Short and Long Answer Questions (60 marks)
2. **All** questions are compulsory
3. Write **ALL** answers in the spaces provided within this paper.
4. This exam is **marked out of 100** and is worth **50%** of your **overall course marks**.
5. To **pass** this paper you require **at least 40% for this exam** which is equivalent to **40 marks out of 100**
6. This paper has 18 pages including this page.

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Section A: Coding Questions 40 Marks

Question 1: ASP.Net

Web forms let you build interactive web pages that collect information from a user and process it on the web server. The Web page in figure 1 below shows an example of a typical web form. Use figure 1 web form and later information to answer the questions that follow.

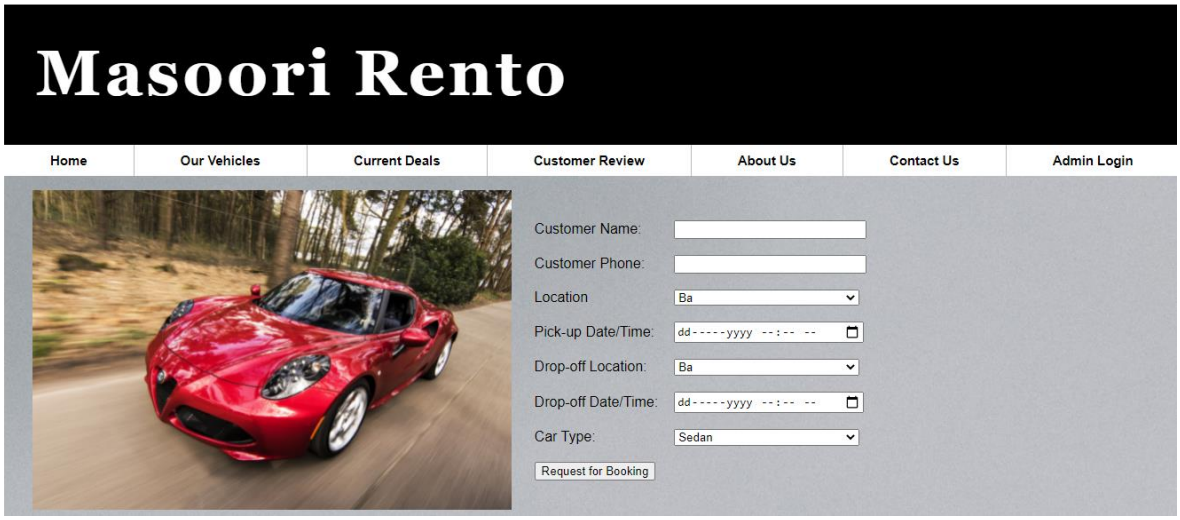

Figure 1: Web Form

a. Write the ASP code for txtCusPhone Textbox. **(2 marks)**

b. Write the ASP code for ddlCarType Dropdown List in asp.net with 2 List items (Sedan and SUV). **(3 marks)**

c. Drop-off Location option has a dropdown list (ddlDropoffLocation). The following are the options in ddlDropoffLocation: Ba, Lami, Lautoka, Nadi, and Navua. This list uses Locations.xml as a data source. Write the XML code for Locations.xml. **(5 marks)**

d. Figure 2 below represent the RentalCar table in Microsoft Access. What control is used to connect the web form to the RentalCar table in Microsoft Access? **(1 mark)**

RentalCar

Field Name	Data Type
ID	AutoNumber
Type	Short Text
Pickup_Location	Short Text
Pickup_DateTime	Date/Time
Dropoff_Location	Short Text
Dropoff_DateTime	Date/Time
Cus_Name	Short Text
Cus_Phone	Short Text

Figure 2: RentalCar Table

e. Write the code to extract the value from the following controls: **(1 mark)**

ddlCarType: _____

txtDropoffDateTime: _____

f. The customer phone number should only accept numeric values and have a total of ten digits.

i. What control should be used to validate txtCusPhone number? **(1 mark)**

ii. Write down the validation expression to make sure the customer does not input any invalid phone number. **(1 mark)**

iii. What control should be used to force the customer to enter a phone number before submitting the form? **(1 mark)**

g. Upon clicking the "Request for Booking"" (btnRequestBooking) button, the customer information will be inserted into the database. Referring to figure 3 write the missing VB code that will handle btnRequestBooking click event. Part of the code of the Event Handler is given below. **(7 marks)**

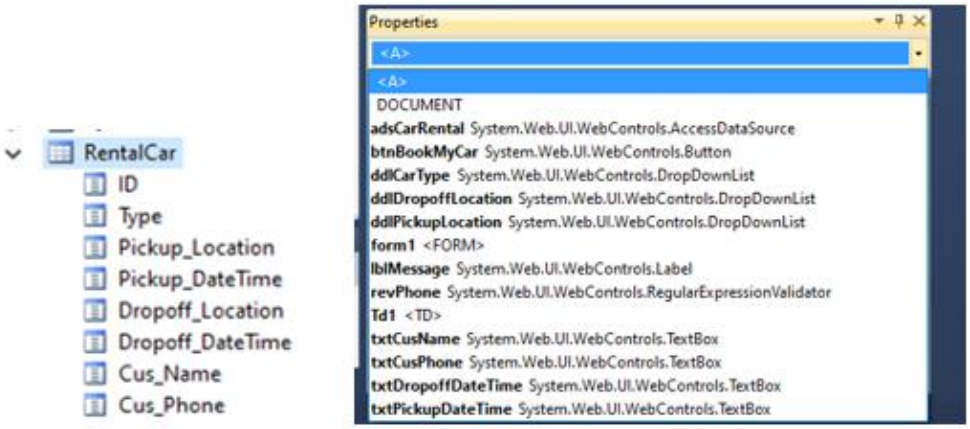

Figure 3

```
Protected Sub btnRequestInfo_Click(sender As Object, e As
System.EventArgs)Handles btnRequestInfo.Click

        ' write the missing VB codes below

End Sub
```

h. Write the ASP code for Masoori Rento to have a printer friendly website by referencing the print.css file that will render successfully when a webpage is printed or viewed in print preview mode. **(1 mark)**

i. Write a CSS style to hide everything that is inside the header tag. **(1 mark)**

j. Masoori Rento wants to make the website mobile friendly (540px and below). Write a media query that can be used to write CSS for mobile. **(1 mark)**

Question 2: Navigation

The navigation below in Figure 4 shows a simple horizontal standard navigation for a website.

a. Write the appropriate HTML codes that will markup the navigation in Figure 4 below to make meaningful semantic navigation. **(5 marks)**

Figure 4

b. Write style rules that can be applied to the navigation in figure 4 by adding the following properties:
   - List style none.
   - Horizontal display.
   - Navigation border: 1px solid #333.
   - Navigation border-bottom: 5px solid #00bff3.
   - Display is block.
   - Background color: #eee.
   - 1.5rem font-size.

(6 marks)

Question 3: Tables

Given the HTML source code for the table below draw the table when it is rendered.

(4 marks)

```
<table border="1">
  <tr>
    <th>Name</th>
    <th>Email address</th>
    <th>Phone Contact</th>
  </tr>
  <tr>
    <td>Racheal</td>
    <td>abc@company.com</td>
    <td>
      <table border=1>
            <tr>
              <th id="nested">Home Phone</th>
              <th id="nested">Work Phone</th>
              <th id="nested">Cell Phone</th>
            </tr>
            <tr>
              <td>9999-880</td>
              <td>9999-881</td>
              <td>9999-882</td>
             </tr>
        </table>
      </td>
  </tr>
</table>
```

Section B: Short/Long Answer Questions 60 Marks

Question 1

Even though an increasingly wide variety of devices support the Web, your primary goal as a Web page designer is to create Web pages for the computer medium.

a. One design principle you can follow is to make your design portable. What does it mean by making your design portable? **(2 marks)**

b. Another design principle is to design for the screen. Your computer monitor, the destination for your Web pages, is different from print-based media. Identify three differences between your computer monitor and print-based media that you must take into account when planning your Web site. **(3 marks)**

Question 2

Uniform Resource Identifiers, also known as Uniform Resource Locator (URL), is the name we give to Internet Addresses. It is the Webs own unique way to differentiate and identify resources on the web. The resources could be documents, images, downloadable files, services, electronic mailboxes, etc. A complete URL is made up of four parts. List and explain each part with an example? **(6 marks)**

Question 3

List four human factors to consider when building a Web site. **(2 marks)**

Question 4

List four technical factors to consider when building a Web site. **(2 marks)**

Question 5

What does the <!DOCTYPE> statement specify? **(2 marks)**

Question 6

You have been hired to develop your local bookstore's website and are required to integrate the following site requirements:

- Logo and mission statement should be featured.
- Search box.
- Links to the other pages of the website should be available.
- Weekly specials should be a main feature content.
- Advertisements of related books, deals and news should also feature.
- Copyright statement should feature as well.
- Uniform look and feel for all the pages, the only difference should be the actual content.

With the aid of a diagram, illustrate how you would design a responsive version of the homepage for mobile devices that have a screen size of 320px. **(10 marks)**

Question 7

With the aid diagrams, discuss the differences between two -tier and three -tier Client/Server Architecture. Your discussion should include the following:

- Elements of two-tier system and their roles/functions
- How information flows in two-tier architecture
- Elements of three-tier system and their roles/functions
- How information flows in three-tier architecture **(8 marks)**

Question 8

The University of the South Pacific (USP) website is very compelling to use, attractive, easy to navigate and a very popular website. In designing the USP website, the designers took into consideration the following pointers:

- Crafting the look and feel
- Design portability
- Easy access to your information
- Planning for Clear presentation of information
- Design for low bandwidth

For each consideration discuss how USP website implemented them. **(10 Marks)**

Question 9

Suppose you are part of the web designing team for the university's website. How would you go about designing and developing USP's website so that the web site becomes live? **(5 marks)**

Question 10

With reference to USP website, how would you design it so that the device you are using to view it determines how best to render this page? **(5 Marks)**

Question 11

With reference to USP website, discuss how you would go about implementing the first two stages of the web development life cycle by identifying key activities and artefacts for each stage.

(5 Marks)

Stage 1_____

Activities

_____
_____
_____
_____
_____
_____
_____

Artefacts

_____
_____
_____

Stage 2_____

Activities

_____
_____
_____
_____
_____
_____
_____

Artefacts

_____
_____
_____

Draft Page

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Draft Page

THE END